

EDUCATION

University of Texas at Austin

Ph.D. in Operations Research, GPA: 4.00/4.00

Austin, TX

Aug 2021 – May 2027

- **Research Methods:** Large-Scale Nonlinear Stochastic Constrained Optimization
- **Research Applications:** Power Systems, Communication and Transportation Networks, Fair AI

University of Texas at Austin

M.S. in Statistics, GPA: 3.97/4.00

Austin, TX

Aug 2021 – Jul 2026

Carnegie Mellon University

M.S. in Energy Science, GPA: 3.98/4.00

Pittsburgh, PA

Jan 2017 – May 2018

Indian Institute of Technology Delhi

B.Tech. in Electrical Engineering (Power), GPA: 8.1/10.0

New Delhi, India

Aug 2012 – May 2016

RESEARCH EXPERIENCE

The University of Texas at Austin

Graduate Research Assistant – Walker Dept. of Mechanical Engineering

Austin, TX

Primary advisor: Prof. Raghu Bollapragada

Aug 2023 – Present

- **BDFO:** Proposed derivative-free optimization framework for bilevel programming. [[Paper: In Prep](#)] [[Code: In Prep](#)]
- **MESCAL:** Proposed and analyzed an augmented Lagrangian framework providing second-order convergence guarantees for nonconvex expectation-constrained optimization. [[Paper](#)] [[Code: On Request](#)]
- **SALSA:** Developed a rate-optimal method for expectation-constrained convex optimization using adaptive sampling strategies. [[Paper: On Request](#)] [[Code: On Request](#)]

Co-advised with Prof. Benjamin D. Leibowicz

- **Camera Placement Optimization:** Developed MIP and adaptive sampling framework to maximize camera visibility coverage (3.3–16.0% improvement). [[Paper](#)] [[Code](#)]

Graduate Research Assistant – McCombs School of Business

Advisors: Prof. Ananth Balakrishnan and Prof. Prakash Mirchandani

Aug 2022 – Dec 2024

- **SDSS (Vehicle Routing Optimization):** Developed network optimization algorithm for vehicle routing problems, yielding > 10% reduction in optimality gap relative to disaggregated resource customer models. [[Paper: In Prep](#)] [[Code: On Request](#)]

Amazon

Research Scientist Intern – FBA Science

Seattle, WA

May 2025 – Aug 2025

- **Seller Behavioral Modeling & Forecasting:** Developed data-driven replenishment behavior framework and deep learning model predicting inbound parameters (reduced weekly sMAPE error from 115% to 79%); presented research insights to FBA leadership.
- **Ecosystem Contract Optimization:** Designed newsvendor-based fee contracts improving customer surplus (+11%), seller profits (+15%), and Amazon revenues (+21%); executive leadership approved model for active pilot testing.

- **Power Systems Optimization & Policy Analysis:** Led carbon capture (45Q incentive) and water usage optimization studies in Pyomo/GAMS—achieving a 30% Texas power plant water reduction and directly briefing DoE leadership on auxiliary cost gaps.
- **Deep Learning for Geological Storage:** Engineered a PyTorch ConvLSTM spatiotemporal model for DoE’s multi-lab SMART Initiative to evaluate long-term geological CO₂ reservoir storage viability.
- **Data Pipelines & Environmental Policy:** Built ETL pipeline analyzing regional power plant effluent data to inform DoE environmental regulatory policy across Northern Appalachian watersheds.

JOURNAL PUBLICATIONS

- [1] Y. Kumar, R. Bollapragada, and B. D. Leibowicz, Efficient mathematical programming formulation and algorithmic framework for optimal camera placement, *Computers & Operations Research*, (2026), 107541. [\[Paper\]](#) [\[Code\]](#)
- [2] I. P. Cabra, S. Sharma, C. Callahan, K. Jani, Y. Kumar, V. Toetz, and E. Shuster, Market analysis for the integration of new power technologies: A case study of the deployment of hybrid fossil-based generator plus energy storage (ES-FE), *Applied Energy*, 351 (2023), 121861. [\[Paper\]](#)
- [3] Y. Kumar, R. Hoesly, A. Venkatesh, E. Shuster, and A. Iyengar, Effects of short-term water constraints on electricity dispatch: a case study of ERCOT and SPP regions, *ACS ES&T Water*, 2 (2022), pp. 749–758. [\[Paper\]](#)

WORKING PAPERS (UNDER REVIEW AND IN PREPARATION)

- [1] R. Bollapragada, and Y. Kumar, Stochastic augmented lagrangian framework with second-order convergence guarantees for nonconvex expectation-constrained optimization, *Mathematics of Operations Research*, (under review). [\[Paper\]](#) [\[Code: On Request\]](#)
- [2] R. Bollapragada, and Y. Kumar, Augmented lagrangian methods for expectation-constrained optimization via adaptive sampling strategies, (manuscript in preparation). [\[Paper: On Request\]](#) [\[Code: On Request\]](#)
- [3] R. Bollapragada, and Y. Kumar, Derivative-free optimization method for bilevel programming, (manuscript in preparation). [\[Paper: In Prep\]](#) [\[Code: In Prep\]](#)

TECHNICAL CONFERENCES & INVITED TALKS

- [1] On Second-order Convergence Guarantees for Expectation-constrained Optimization Problems, November 2026. INFORMS Annual Meeting, San Francisco, CA.
- [2] Adaptive Sampling Methods for Expectation-Constrained Stochastic Optimization, March 2026. INFORMS Optimization Society Conference, Atlanta, GA.
- [3] Data-driven Seller Replenishment Behavior Framework, August 2025. Amazon FBA Science Review Meeting, Seattle, WA.
- [4] Efficient Mathematical Programming Formulation and Algorithmic Framework for Optimal Camera Placement, October 2024. INFORMS Annual Meeting, Seattle, WA.
- [5] Resource Planning and Equitable Work Assignment for On-Site Services, March 2024. International Network Optimization Conference, Dublin, Ireland.
- [6] Neural Network-Based Surrogate Models for Joint Prediction of Reservoir Pressure and CO₂ Saturation, October 2020. SMART Initiative: Virtual Annual Review Meeting.
- [7] Economics of Flexible Carbon Capture and Storage (CCS) Systems, September 2020. International Coal Conference, Pittsburgh, PA.
- [8] Electricity Dispatch of Thermoelectric Power Generation under Short-Term Water Consumption Constraints: ERCOT Case Study, November 2019. International Water Conference, Orlando, FL.
- [9] Power Generation under Short-Term Water Constraints, July 2019. ASME Power Conference, Salt Lake City, UT.

- [10] Appalachian Energy Water Landscape: Key Emerging Issues, July 2019. ASME Power Conference, Salt Lake City, UT.

HONORS, AWARDS & FUNDING

- **Outstanding Graduate Research Fellowship**, UT Austin *Jun 2026 – May 2027*
Awarded full tuition and stipend for outstanding academic and research excellence.
- **Professional Development Award**, UT Austin *2024*
Competitive travel grant awarded for research presentation at the 2024 INFORMS Annual Meeting.
- **Graduate Engineering Council Travel Grant**, UT Austin *2024*
Awarded for research presentation at the 2024 International Network Optimization Conference.
- **Spotlight Award**, KeyLogic Systems *Sep 2019*
Awarded for Texas short-term power systems water shortage analysis.
- **Spotlight Award**, KeyLogic Systems *May 2019*
Recognized for Northern Appalachian water landscape study.
- **Course Assistant Award**, Carnegie Mellon University *2017 – 2018*
Recognized for exceptional pedagogical support and service to university faculty.
- **Summer Undergraduate Research Award**, IIT Delhi *2013*
Secured competitive funding and boarding for independent summer research.
- **All India Rank 69 (Top 0.005%)**, All India Engineering Entrance Examination (AIEEE) *2012*
Ranked 69th nationwide out of 1.2 million participating students.

TEACHING EXPERIENCE

- The University of Texas at Austin** Austin, TX
Graduate Teaching Assistant
- Teaching Assistant, *Engineering Finance*, Walker Dept. of Mechanical Engineering *Fall 2025*
 - Teaching Assistant, *Ethics in Artificial Intelligence*, Dept. of Computer Science *Fall 2022*
 - Teaching Assistant, *Probability and Statistics for Civil Engineers*, Dept. of Civil Engineering *Fall 2021*
- Carnegie Mellon University** Pittsburgh, PA
Graduate Course Assistant & Tutor
- Course Assistant, *Energy Demand and Utilization*, College of Engineering *Spring 2018*
 - Course Assistant, *Energy Policy and Economics*, College of Engineering *Spring 2018*
 - Peer Tutor, *Introduction to Machine Learning*, School of Computer Science *Spring 2018*
 - Course Assistant, *Emerging Energy Policies*, College of Engineering *Fall 2017*
 - Course Assistant, *Energy Innovation and Entrepreneurship*, College of Engineering *Fall 2017*

PROFESSIONAL SERVICE & LEADERSHIP

Conference & Journal Service

- Session Chair (×3), *Advances in Nonlinear Continuous Optimization I, II, III*, INFORMS Annual Meeting *2026*
- Reviewer, *INFORMS Optimization Society Conference* *2026*
- Judge, *Technology & Science Undergraduate Research Forum* *2026*

Professional Memberships & Student Leadership

- Member, *Institute for Operations Research and the Management Sciences (INFORMS)* *2023 – Present*
- Treasurer, *INFORMS Student Chapter*, UT Austin *2025 – 2027*
- Quizzing Club Representative, *Indian Institute of Technology Delhi* *2013 – 2014*

ADDITIONAL PROFESSIONAL AND RESEARCH EXPERIENCE

UT Energy Institute

Austin, TX

Graduate Research Assistant

Aug 2021 – May 2022

- **VPP Grid Integration:** Implemented economic dispatch and unit commitment MIP models for vehicle-to-grid integration in virtual power plant; evaluated utility tariff structures for locational feasibility. [[Code: On Request](#)]

Catalyst Cooperative

Pittsburgh, PA (Remote)

Data Scientist

Apr 2020 – Oct 2020

- **Public Utility Data Liberation (PUDL):** Developed open-source spatiotemporal energy demand allocation models and automated ETL pipelines integrating FERC-714, EIA, NREL, and US Census datasets. [[Codebase](#)]

Nikola Power

Denver, CO

Software Engineering Intern

Sep 2018 – Nov 2018

- **Load & Generation Forecasting:** Built CNN and LSTM deep neural network architectures for day-ahead load forecasting (MAPE < 12%) and solar generation forecasting (MASE < 0.1).

Carnegie Mellon University

Pittsburgh, PA

Research Engineer

Jul 2018 – Oct 2018

- **Power Sector Carbon Index:** Developed state-level emissions indices analyzing 17 years of US power sector data; categorized emissions profile across NERC and ISO regions. [[Codebase](#)]

COMMUNITY OUTREACH

- **Audiobook Reviewer**, Texas Talking Book Program *2025 – Present*
- **Volunteer**, Inside Books Project, Austin *2025 – Present*
- **Developer**, Pittsburgh Code for Good *2019 – 2020*